STUDENT PERFORMANCE CASE STUDY

A clear review of the notebook, its results, and what they tell us

Average marks	**74.55**
Students scoring above 75	**10**
Passes / fails	**18 / 2**
Pass rate	**90%**
Attendance–marks correlation	**–0.52**

1. Introduction and Summary

This case study looks at student marks and attendance using Python. The notebook calculates the average mark, counts students scoring above 75, checks the relationship between attendance and marks, and separates students into pass and fail groups. It also creates a scatter plot and a histogram to make the results easier to understand.

The saved results show an average mark of 74.55. Ten students scored above 75, and the pass/fail count is 18 to 2, using 50 marks as the pass threshold. The notebook also reports a correlation of –0.52 between attendance and marks. That negative value needs careful interpretation; it does not prove that attending more classes leads to lower marks.

2. How the Notebook Works



In simple terms, the notebook reads the dataset, calculates summary values, identifies students above the selected marks threshold, calculates correlation, and counts pass and fail results. It then plots attendance against marks and shows how marks are distributed.

3. What the Results Mean

3.1 Average mark: 74.55

The average gives a quick idea of overall performance, but it does not tell the whole story. Two classes can have the same average while having very different mark distributions. The median, standard deviation, and minimum and maximum marks would help show how spread out the results are.

3.2 Students scoring above 75: 10

The notebook uses a strict 'greater than 75' condition. This means a mark of exactly 75 is not counted in this group. If there are 20 students in the dataset, as the pass/fail totals suggest, then 10 out of 20 students—half the group—scored above 75. This assumes all rows were classified and no marks were missing.

3.3 Passes and fails: 18 and 2

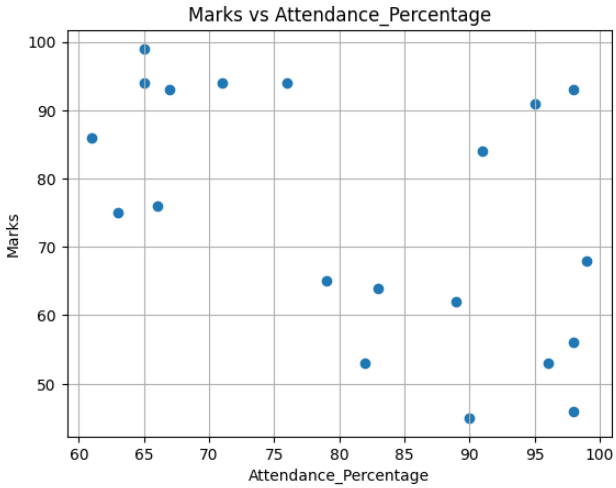
The notebook treats marks of 50 or more as a pass. Based on the saved counts, 18 students passed and 2 failed, giving a pass rate of 90% if the dataset contains 20 fully classified records. The pass threshold should match the rule used by the institution or study.

3.4 Attendance and marks: correlation −0.52

The notebook reports a Pearson correlation of −0.52. In this dataset, attendance and marks move in opposite directions to a moderate degree: higher attendance values tend to appear alongside lower marks. This is an association, not proof of cause and effect. Before drawing conclusions, check for unusual records, incorrect values, small-sample effects, or a column that may have been labelled incorrectly. The missing CSV prevents checking those possibilities here.

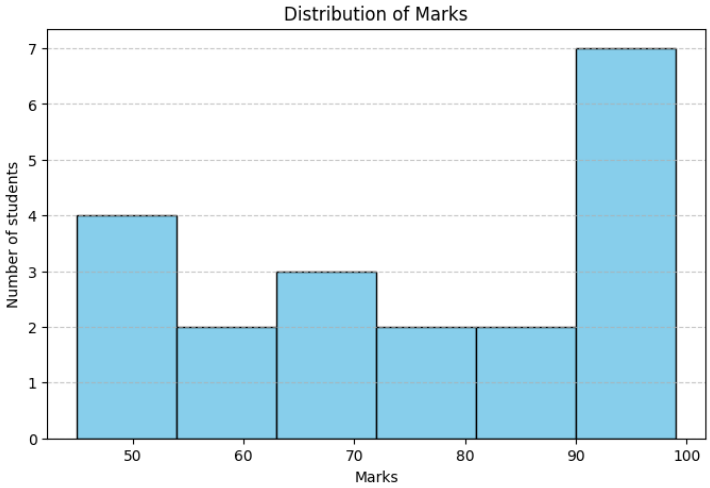
4. Charts and Visual Findings

4.1 Attendance compared with marks



The scatter plot places attendance on one axis and marks on the other. It helps show whether students with different attendance levels tend to receive different marks. Look for isolated points, clusters, or patterns that do not follow a straight line. The chart should be checked alongside the reported correlation, not used on its own to claim that attendance causes a change in marks.

4.2 Distribution of marks



The histogram groups marks into ranges, making it easier to see where scores are concentrated. The notebook uses six bins. To describe the distribution confidently, compare the chart with the median and spread of marks; the average alone is not enough.

5. Review of the Code

The notebook is a straightforward introduction to pandas and Matplotlib. It covers useful basics, but a few changes would make it more reliable:

• **File path:** The code expects the CSV at `/content/student_performance_dataset.csv`. That path may need to change when running the notebook on another computer.
• **Repeated pass calculation:** The notebook creates the Pass column twice. The second assignment repeats the first calculation, so one is enough.
• **Input checks:** The code assumes the required columns exist and contain numeric values. It does not check missing

data, duplicate rows, or marks and attendance values outside expected ranges.
• **Charts:** Axis labels and titles are included, but a sample size and a trend line could make the plot easier to interpret.

6. Original Notebook Code

The code below keeps the notebook's original logic, including the repeated Pass assignment, so the report reflects the uploaded source rather than silently rewriting it.

Cell 1 — Load data and calculate average

```
import pandas as pd
df = pd.read_csv('/content/student_performance_dataset.csv')
avg_marks = df['Marks'].mean()
print(f'Average marks: {avg_marks:.2f}')
```

Cell 2 — Count marks above 75

```
high_performance = df[df['Marks'] > 75]
high_performance_count = len(high_performance)
print(f'Number of students with marks above 75: {high_performance_count}')
```

Cell 3 — Plot attendance against marks

```
import matplotlib.pyplot as plt
plt.scatter(df['Attendance_Percentage'], df['Marks'])
plt.xlabel('Attendance_Percentage')
plt.ylabel('Marks')
plt.title('Marks vs Attendance_Percentage')
plt.grid(True)
```

Cell 4 — Calculate correlation

```
correlation = df['Attendance_Percentage'].corr(df['Marks'])
print(f'Correlation between Attendance_Percentage and Marks: {correlation:.2f}')
```

Cell 5 — Create pass flag

```
df['Pass'] = df['Marks'] >= 50
```

Cell 6 — Recalculate pass flag (duplicate step)

```
df['Pass'] = df['Marks'].apply(lambda x: True if x >= 50 else False)
```

Cell 7 — Count results and plot marks

```
passes = df['Pass'].sum()
fails = (~df['Pass']).sum()
print(f'Passes: {passes}')
print(f'Fails: {fails}')
plt.figure(figsize=(8,5))
plt.hist(df['Marks'], bins=6, edgecolor='black', color='skyblue')
plt.xlabel('Marks')
plt.ylabel('Number of students')
plt.title('Distribution of Marks')
plt.grid(True, axis='y', linestyle='--', alpha=0.7)
plt.show()
```

7. Conclusion

This notebook gives a basic overview of student marks and attendance. The saved outputs show an average mark of 74.55, 18 passes and 2 fails, and 10 students scoring above 75. The reported attendance–marks correlation is negative, but the reason for that pattern is not clear from the notebook alone. The next step is to obtain the original CSV, check the data for errors, and rerun the analysis before making any firm conclusions.